

Week 6 Office Hour Notes

6.0 Errata

10/19/24 1:30pm. Full proof of compactness Theorem added, as solutions to Homework 6 have been released.

6.1 Homework 5 Solutions

Omitted to not reveal homework solutions...

6.2 Compactness Theorem

IMPORTANT NOTE: I have written up some much more detailed notes (containing proofs of the crucial lemmas used in the main proof of Compactness Theorem), but because much of this proof is left as an exercise for the Homework 6 due on Monday Oct. 7, I will keep those details concealed until I do my Week 7 Office Hour lecture.

Let Σ be a set of wff's. Recall that we call Σ **satisfiable** if there exists a truth assignment mapping $\bar{v} : \bar{S} \rightarrow \{T, F\}$ such that $\bar{v}(\alpha) = T$, for all $\alpha \in \Sigma$.

Definition 6.2.1. Let

- (i) Σ is **finitely satisfiable** if every finite $\Sigma_0 \subset \Sigma$ is satisfiable, i.e., for every finite $\Sigma_0 \subset \Sigma$, there exists a truth assignment mapping $\bar{v} : \bar{S} \rightarrow \{T, F\}$ such that $\bar{v}(\alpha) = T$, for all $\alpha \in \Sigma_0$.
- (ii) Σ is **maximally finitely satisfiable** if it is finitely satisfiable and for any wff σ , we have $\sigma \in \Sigma$ or $\neg\sigma \in \Sigma$.

Theorem 6.2.2. (Compactness Theorem) Σ is satisfiable iff Σ is finitely satisfiable.

Corollary 6.2.3. If $\Sigma \models \tau$, for some wff τ , then $\Sigma_0 \models \tau$, for some finite $\Sigma_0 \subset \Sigma$.

Proof. Done in **Corollary 17A** in page 60 of *Enderton*. Left to reader to reinvent it. $\square$

Lemma 6.2.4. (the Lindenbaun Lemma) If Σ is finitely satisfiable, then there exists $\Delta \supset \Sigma$ such that Δ is maximally finitely satisfiable.

Proof. Let $\sigma_1, \sigma_2, \dots$ be an enumeration of all wff's. Then define recursively $\Delta_0 := \Sigma$ and for $n \geq 0$, define

$$\Delta_{n+1} := \begin{cases} \Delta_n \cup \{\sigma_{n+1}\} & \text{if } \Delta_n \cup \{\sigma_{n+1}\} \text{ is finitely satisfiable,} \\ \Delta_n \cup \{\neg\sigma_{n+1}\} & \text{otherwise.} \end{cases} \quad (6.1)$$

First, we show that Δ_n , for every $n \geq 0$ is finitely satisfiable by induction on n . For $n = 0$, we are done since $\Delta_0 = \Sigma$ is finitely satisfiable by hypothesis. In the inductive step, we assume Δ_n is finitely satisfiable. In the case where $\Delta_n \cup \{\sigma_{n+1}\}$ is finitely satisfiable, then $\Delta_{n+1} = \Delta_n \cup \{\sigma_{n+1}\}$ is finitely satisfiable and we are done. Otherwise, assume $\Delta_n \cup \{\sigma_{n+1}\}$ isn't finitely satisfiable, and choose finite $\Sigma_0 \subset \Delta_n \cup \{\sigma_{n+1}\}$ that isn't satisfiable. In this case, we have $\Delta_{n+1} = \Delta_n \cup \{\neg\sigma_{n+1}\}$, which we want to show is finitely satisfiable.

Suppose towards contradiction $\Delta_{n+1} = \Delta_n \cup \{\neg\sigma_{n+1}\}$ isn't finitely satisfiable. Choose finite $\Sigma_0' \subset \Delta_{n+1}$ and finite that isn't satisfiable. Note by inductive hypothesis that since $(\Sigma_0 \cup \Sigma_0') \setminus \{\sigma_{n+1}, \neg\sigma_{n+1}\}$ is a finite subset of Δ_n , it is satisfiable, so we can choose $\bar{v} : \bar{S} \rightarrow \{T, F\}$ satisfying $(\Sigma_0 \cup \Sigma_0') \setminus \{\sigma_{n+1}, \neg\sigma_{n+1}\}$. We then have two cases.

Case 1. Suppose $\bar{v}(\sigma_{n+1}) = T$. Then $\bar{v}$ satisfies $((\Sigma_0 \cup \Sigma_0') \setminus \{\sigma_{n+1}, \neg\sigma_{n+1}\}) \cup \{\sigma_{n+1}\} \supset \Sigma_0$, hence $\bar{v}$ satisfies Σ_0 , contradicting the hypothesis that Σ_0 is unsatisfiable.

Case 2. Suppose $\bar{v}(\sigma_{n+1}) = F$. Then since $\bar{v}(\neg\sigma_{n+1}) = T$, we find that $\bar{v}$ satisfies $((\Sigma_0 \cup \Sigma_0') \setminus \{\sigma_{n+1}, \neg\sigma_{n+1}\}) \cup \{\neg\sigma_{n+1}\} \supset \Sigma_0'$, hence $\bar{v}$ satisfies Σ_0' , contradicting the hypothesis that Σ_0' is unsatisfiable.

In both cases, we have shown a contradiction, and we conclude that Δ_{n+1} , in the case where $\Delta_n \cup \{\sigma_{n+1}\}$ isn't finitely satisfiable, is finitely satisfiable, which completes the inductive step of the argument that Δ_n is finitely satisfiable, for every $n \geq 0$.

Finally, we show that $\Delta := \bigcup_{n=0}^{\infty} \Delta_n$ is our desired maximally finitely satisfiable set. First, note that given a finite $\Sigma_0 := \{\tau_1, \dots, \tau_N\} \subset \Delta$, we find that there exists $n \geq 0$ sufficiently large such that $\tau_k \in \Delta_n$, for each $1 \leq k \leq N$ (since $m \leq n$ implies $\Delta_m \subset \Delta_n$), and hence we conclude that Σ_0 as a finite subset of Δ_n is finitely satisfiable, which in turn shows that Δ is finitely satisfiable. Next, take a wff σ and note that σ is σ_n , for some $n \in \mathbb{N}$, and it follows by (6.1) that if $\Delta_{n-1} \cup \{\sigma_n\}$ is finitely satisfiable, then $\sigma := \sigma_n \in \Delta_n = \Delta_{n-1} \cup \{\sigma_n\} \subset \Delta$, and otherwise we have $\neg\sigma := \neg\sigma_n \in \Delta_n = \Delta_{n-1} \cup \{\neg\sigma_{n+1}\} \subset \Delta$. Either way, the maximal finite

satisfiability condition is satisfied, and the proof of the *Lindenbaun Lemma* is complete. $\square$

Lemma 6.2.5. If Δ is a maximally finitely satisfiable set, then $v_\Delta : \bar{S} \rightarrow \{T, F\}$ defined by

$$v_\Delta(\sigma) = \begin{cases} T & \text{if } \sigma \in \Delta \\ F & \text{otherwise} \end{cases} \quad (6.1)$$

is a well-defined truth assignment mapping.

Proof. Follows immediately by **Question 2 of Homework 6** since we can define $v : S \rightarrow \{T, F\}$ by $v(A) = T \iff A \in \Delta$ found in **Section 7.1** of the [Week 7 Office Hour Notes](#), and we can define $v_\Delta := \bar{v}$, since we have shown in this homework question that $\bar{v}(\sigma) = T \iff \sigma \in \Delta$. $\square$

Finally, we are ready to prove the compactness theorem.

Proof of the Compactness Theorem (Theorem 6.2.2).

$\implies$ If Σ is satisfiable, we can choose truth assignment mapping $v : \bar{S} \rightarrow \{T, F\}$ satisfying Σ , and it follows that for every finite $\Sigma_0 \subset \Sigma$, we find v also satisfies Σ_0 ; therefore, we find Σ is finitely satisfiable.

$\impliedby$ Conversely, suppose Σ is finitely satisfiable. By the *Lindenbaun Lemma* (**Lemma 6.2.4**), we can choose a maximal finitely satisfiable set $\Delta \supset \Sigma$. Then by **Lemma 6.2.5**, we can define truth assignment mapping $v_\Delta : \bar{S} \rightarrow \{T, F\}$ by (6.1), which we find satisfies Σ , since for all $\sigma \in \Sigma \subset \Delta$, we have $v(\sigma) = T$, $\square$